

Dynamic trajectory stability as a generalization criterion: Lyapunov's approach to the analysis of stochastic learning

Running title: Generalization via trajectory stability

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Abstract

We consider the problem of controlling the expected generalization error of stochastic learning algorithms from the perspective of dynamical stability. Unlike classical approaches that rely on the complexity of the hypothesis class or on the stability of a single final model, we study the entire law of the iterates (logits, probabilities, or embeddings) after they enter a local basin of the loss landscape. Our analysis is based on a stochastic Lyapunov drift condition that ensures contraction of the trajectory and bounds its sensitivity to perturbations of the training data. Using martingale concentration inequalities and Wasserstein geometry, we derive non-asymptotic probabilistic bounds on the distance between the trajectory distributions corresponding to neighboring datasets. These bounds are then translated into explicit control of the generalization gap by means of the classical algorithmic stability theorem of Bousquet and Elisseeff. The main result is an upper bound on the expected generalization gap that decomposes into three terms: a data-sensitivity term that decays with the sample size, an exponentially decaying term due to initial conditions, and a barrier term that penalizes exits from the basin. This provides a principled connection between the dynamical properties of the optimization process and the model's ability to generalize, complementing static complexity measures and endpoint stability analyses.

Keywords. Stochastic gradient descent, algorithmic stability, generalization bounds, Lyapunov drift, trajectory laws, Wasserstein distance

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1. Introduction

Understanding why overparameterized neural networks trained by stochastic gradient descent (SGD) generalize well despite having sufficient capacity to memorize arbitrary data remains one of the central open problems in deep learning theory (Zhang et al., 2017, Belkin et al., 2019). Classical statistical learning theory, based on complexity measures such as VC-dimension or Rademacher complexity, often yields vacuous bounds for modern deep architectures. This has motivated a shift toward algorithm-dependent analyses that exploit the properties of the optimization dynamics rather than the structure of the hypothesis space alone.

A line of work in this direction is algorithmic stability, pioneered by Bousquet (Bousquet et al., 2020). The key insight is that if a learning algorithm's output does not change much when a single training example is removed or replaced, then the algorithm generalizes well. This notion has been extended to uniformly stable algorithms and shown to yield high-probability generalization bounds. However, classical stability bounds often remain loose for deep learning, and the analysis typically focuses on the final model rather than the entire learning dynamics. Other influential works on stability include (Devroye & Wagner, 1979; Kearns & Ron, 1999; Hardt et al., 2016), which established connections between leave-one-out stability, generalization, and the dynamics of SGD.

A complementary perspective studies the loss landscape of neural networks. It is now well established that in overparameterized regimes, global minimizers form connected manifolds rather than isolated points (Nguyen, 2023), and the loss landscape can be described as a collection of percolating wide valleys. The flatness of the basin surrounding a trained solution has been widely discussed as a heuristic proxy for generalization: flatter minima are believed to correspond to models that are robust to parameter perturbations. This intuition has been formalized in the context of high-dimensional Gaussian mixtures by (Baldassi et al., 2020), where wide flat minima are shown to correlate with optimal generalization.

However, naive sharpness measures are sensitive to parameterization, and the relationship between geometric flatness and generalization is subtle. Recent work has also uncovered the emergence of expansive stability regions ("basins") in the loss landscape of large language models, where models exhibit nearly identical performance within the basin

but collapse outside it. These observations highlight the importance of understanding the dynamical stability of the training process.

The dynamics of SGD themselves have been the subject of intense study. A large body of work has analyzed convergence properties under various assumptions, including convexity, Polyak–Łojasiewicz conditions, and more general dissipativity conditions. Recent advances have established connections between gradient flow on population loss and SGD convergence through converse Lyapunov theorems, showing the existence of Lyapunov potentials under mild assumptions. Complementing this, (Chen et al., 2024) developed a Lyapunov theory for finite-sample guarantees of Markovian stochastic approximation, providing a rigorous framework for analyzing the convergence of SGD in non-stationary settings. These results suggest that Lyapunov-based methods provide a unifying framework for analyzing stochastic optimization algorithms.

Wasserstein geometry and transportation inequalities have emerged as powerful tools for studying the stability and concentration of stochastic algorithms. Uniform-in-time Wasserstein stability bounds have been derived for SGD under strongly convex and certain non-convex losses, providing time-uniform guarantees that do not grow with the number of iterations. In a related direction, (Guillin et al., 2026) established uniform-in-time concentration bounds in Wasserstein distance for two-layer neural networks, quantifying the discrepancy between SGD-trained models and their mean-field limits. These bounds rely on developing appropriate Lyapunov functions and illustrate the importance of ergodicity in obtaining time-uniform results. Furthermore, concentration bounds in Wasserstein distance have been proven for two-layer neural networks, quantifying the discrepancy between SGD-trained models and their mean-field limits.

Large deviations theory provides a principled framework for studying rare events and high-probability bounds for stochastic processes. Recent work has applied large deviations to SGD, obtaining upper bounds for strongly convex functions that capture analytical dependence on the noise distribution and problem parameters, going beyond conventional mean-square error analysis. Specifically, (Bajovic et al., 2022) derived large deviations rates for SGD with strongly convex losses, providing exponential decay bounds that depend explicitly on the noise structure. In the non-convex setting, (Raginsky et al., 2017) analyzed SGD through the lens of stochastic gradient Langevin dynamics, establishing non-asymptotic convergence and large-deviation properties. In the context of deep learning, heavy-tailed large deviations have been used to characterize the global dynamics of SGDs and their ability to avoid sharp local minima, which are believed to lead to poor generalization. The classical Freidlin–Wentzell theory of large deviations for stochastic dynamical systems provides a framework for estimating exit probabilities from domains of attraction. Martingale concentration inequalities have long been a staple of stochastic analysis. The Azuma–Hoeffding inequality and its extensions provide non-asymptotic control of the supremum of martingales with bounded differences. In the context of stochastic optimization, these tools have been used to derive concentration bounds for the iterates of SGD under various noise assumptions. The combination of martingale techniques with Lyapunov drift arguments is particularly powerful, allowing for the derivation of exponential concentration bounds that depend on the contraction rate of the dynamics.

Output-space analysis offers a way to circumvent the difficulties associated with parameter-space symmetries in neural networks. Permutation symmetries, scaling invariances, and other redundancies make distances in parameter space poorly suited for comparing models. Analyzing the trajectories of model outputs—such as logits, probabilities, or embeddings—provides a more meaningful and invariant notion of proximity. This perspective aligns with recent work on learning trajectories as generalization indicators, which has shown that the evolution of training dynamics encodes information about final generalization performance. In particular, (Fu et al., 2023) demonstrated that the entire learning trajectory, rather than just the final model, contains predictive signals about generalization, supporting the trajectory-centric approach adopted in our work.

Despite these advances, a framework that combines algorithmic stability, Lyapunov drift, concentration inequalities, Wasserstein geometry, and large deviations to analyze the generalization of trajectory laws—rather than final models—has been lacking. Existing stability analyses typically consider the sensitivity of the final output to data perturbations, ignoring the information contained in the full training trajectory. Concentration bounds for SGD often focus on parameter-space convergence or prediction error at a fixed time, without providing a direct link to generalization through stability. Large deviations results for exit probabilities from basins have not been systematically connected to generalization bounds. Recently, Timofeev (Timofeev, 2026) explored an alternative route to generalization via explicit entropy maximization of embeddings, showing that it reduces the Rademacher complexity of the hypothesis class. Our trajectory-based framework complements this line of work by analyzing the stability of the learning dynamics rather than the properties of the final representation.

In this work, we propose a framework that addresses these gaps by analyzing the entire stochastic learning trajectory after it enters a local basin. We analyze the entire stochastic learning trajectory after it enters a local basin, studying the law of the trajectory rather than a single final model. This trajectory law is mapped to the output space (logits, probabilities, embeddings), avoiding the pathologies of parameter-space metrics. We introduce three key quantities:

1. **Basin retention probability** $\mathcal{B}_T = 1 - \mathbb{P}(\tau_B \leq T)$, measuring the dynamical stability of the learning mode.
2. **Output trajectory concentration index** $\mathcal{C}_{\varepsilon, T}$, quantifying the spread of output trajectories around their mean.

3. **Neighboring-data trajectory stability** $\mathcal{S}_T = \sup_i W_1(v_{S,X}^{t_0:T}, v_{S^{(i)},X}^{t_0:T})$, measuring the sensitivity of the trajectory distribution to replacing a single training example.

Our main theoretical result establishes a non-asymptotic bound on the expected generalization gap in terms of these quantities:

$$\mathbb{E}[R(\mathcal{A}(S)) - \hat{R}_S(\mathcal{A}(S))] \leq \frac{L_\ell L_\Phi C}{n} + C_{\text{burn}} e^{-\alpha(t_0 - t_{\text{init}})} + C_{\text{exit}} \exp\left(-\frac{R^2(1 - (1 - \alpha)^2)}{2\eta^2 B^2(T - t_0 + 1)}\right).$$

This bound is derived through a chain of arguments:

- Lyapunov drift controls the contraction of trajectories and bounds their sensitivity to data perturbations.
- Martingale concentration inequalities provide non-asymptotic probabilistic control of deviations.
- Wasserstein distance between trajectory laws for neighbouring datasets quantifies algorithmic stability at the trajectory level.
- Classical algorithmic stability theory (Bousquet–Elisseeff) (Bousquet & Elisseeff, 2000, 2002) translates trajectory stability into generalization bounds.

Our framework thus provides a connection between the dynamical properties of the optimization process and the generalization capability of the learned model. Our framework combines these diverse mathematical tools to provide a unified perspective on generalization through the lens of trajectory-law stability. This approach bridges the dynamical analysis of optimization with classical stability-based generalization theory.

The paper is organized as follows. In Section 2, we introduce the stochastic learning dynamics and define the basin, trajectory laws, and output-space mapping. Section 3 presents our main results: concentration of output trajectories (Proposition 1), probability of exiting the basin (Proposition 2), and stability of trajectory distributions under data replacement (Proposition 3). In Section 4, we combine these results to prove our main theorem on the generalization gap. Section 5 discusses the limitations of our approach and directions for future work.

Main Results

Proposition 1 (Concentration of output trajectories inside a basin) Assumptions.

Let the following simplifying assumptions hold.

1. **Parameter space** is Euclidean: $\Theta \subseteq \mathbb{R}^d$.
2. **Output map** $\Phi: \Theta \rightarrow \mathcal{Y}$ is L_Φ -Lipschitz with respect to the norm $\|\cdot\|$:

$$\|\Phi(\theta) - \Phi(\theta')\| \leq L_\Phi \|\theta - \theta'\|, \forall \theta, \theta' \in \Theta.$$

3. **Stochastic dynamics** is given by

$$\Theta_{t+1} = \Theta_t - \eta \nabla \ell(\Theta_t, \xi_t),$$

where $\eta > 0$ is a constant step size and ξ_t is independent noise.

4. **Conditional expectation** of the update is a strict contraction with respect to a point θ^* :

$$\mathbb{E}[\Theta_{t+1} | \mathcal{F}_t] = \theta^* + (1 - \alpha)(\Theta_t - \theta^*), \alpha \in (0, 1).$$

Equivalently, $\mathbb{E}[\Theta_{t+1} - \theta^* | \mathcal{F}_t] = (1 - \alpha)(\Theta_t - \theta^*)$.

5. **Martingale difference** (noise term)

$$M_{t+1} := \Theta_{t+1} - \mathbb{E}[\Theta_{t+1} | \mathcal{F}_t]$$

is bounded almost surely: $\|M_{t+1}\| \leq B$ a.s. for all t .

6. **Burn-in** t_0 is chosen such that the initial deviation is negligible: $\Theta_{t_0} = \theta^*$ (or, in general, a decaying term is handled separately).

7. **Basin** $B \subset \Theta$ is defined as a ball of radius R in parameter space:

$$B = \{\theta \in \Theta: \|\theta - \theta^*\| \leq R\},$$

inside which the dynamics satisfies Assumptions 4–5. We consider the event $\{\tau_B > T\}$, where $\tau_B = \inf\{t \geq t_0: \Theta_t \notin B\}$.

Statement. For any $\varepsilon > 0$, the following inequality holds:

$$\mathbb{P}\left(\sup_{t_0 \leq t \leq T} \|\Phi(\Theta_t) - \mathbb{E}[\Phi(\Theta_t)]\| > \varepsilon\right) \leq 2 \exp\left(-\frac{\varepsilon^2(1 - (1 - \alpha)^2)}{2L_\Phi^2 B^2 \eta^2 (T - t_0 + 1)}\right) + \mathbb{P}(\tau_B \leq T).$$

Proof.

Step 1: Centering the process.

Let $\mu_t := \mathbb{E}[\Theta_t]$. Taking expectation in Assumption 4, we get

$$\mu_{t+1} - \theta^* = (1 - \alpha)(\mu_t - \theta^*),$$

hence $\mu_t \rightarrow \theta^*$. Define the centered variable

$$Z_t := \Theta_t - \mu_t.$$

Using Assumption 4,

$$\mathbb{E}[\Theta_{t+1} | \mathcal{F}_t] - \mu_{t+1} = [\theta^* + (1 - \alpha)(\Theta_t - \theta^*)] - [\theta^* + (1 - \alpha)(\mu_t - \theta^*)] = (1 - \alpha)(\Theta_t - \mu_t) = (1 - \alpha)Z_t.$$

Therefore,

$$Z_{t+1} = (1 - \alpha)Z_t + M_{t+1},$$

where M_{t+1} is a martingale difference with $\|M_{t+1}\| \leq B$.

Step 2: Time decomposition.

Unrolling the recurrence for $t \geq t_0$, we obtain

$$Z_t = (1 - \alpha)^{t-t_0} Z_{t_0} + \eta \sum_{k=t_0}^{t-1} (1 - \alpha)^{t-1-k} M_{k+1}.$$

By Assumption 6, $Z_{t_0} = 0$. Thus,

$$Z_t = \eta \sum_{k=t_0}^{t-1} (1 - \alpha)^{t-1-k} M_{k+1}.$$

Step 3: Bounding the supremum of a weighted sum of martingale differences.

For each fixed t , Z_t is a sum of martingale differences with weights $w_{t,k} = \eta(1 - \alpha)^{t-1-k}$. From the bound $\|M_{k+1}\| \leq B$, the sequence $S_t = Z_t$ (as a function of t) is a martingale (after suitable filtration) with increments bounded in norm. Applying the vector-valued version of the Azuma–Hoeffding inequality for martingale maxima (Freedman, 1975; Dzhaparidze & van Zanten, 2001), we get

$$\mathbb{P}\left(\sup_{t_0 \leq t \leq T} \|Z_t\| > \delta\right) \leq 2 \exp\left(-\frac{\delta^2}{2\eta^2 B^2 \sum_{k=t_0}^T \sum_{t=k}^T (1 - \alpha)^{2(t-1-k)}}\right).$$

We estimate the double sum from above. The inner sum is a geometric progression:

$$\sum_{t=k}^T (1 - \alpha)^{2(t-1-k)} \leq \sum_{j=0}^{\infty} (1 - \alpha)^{2j} = \frac{1}{1 - (1 - \alpha)^2}.$$

Hence,

$$\sum_{k=t_0}^T \sum_{t=k}^T (1-\alpha)^{2(t-1-k)} \leq \frac{T-t_0+1}{1-(1-\alpha)^2}.$$

Substituting this into the previous inequality yields

$$\mathbb{P} \left(\sup_{t_0 \leq t \leq T} \|Z_t\| > \delta \right) \leq 2 \exp \left(-\frac{\delta^2(1-(1-\alpha)^2)}{2\eta^2 B^2(T-t_0+1)} \right).$$

Step 4: Passing to the output space.

From the Lipschitz continuity of Φ (Assumption 2) and the fact that $\mathbb{E}[\Phi(\Theta_t)]$ is the center (since $\mu_t = \mathbb{E}[\Theta_t]$), it follows that

$$\sup_{t_0 \leq t \leq T} \|\Phi(\Theta_t) - \mathbb{E}[\Phi(\Theta_t)]\| \leq L_\Phi \sup_{t_0 \leq t \leq T} \|Z_t\|.$$

Therefore, for any $\varepsilon > 0$,

$$\mathbb{P} \left(\sup_{t_0 \leq t \leq T} \|\Phi(\Theta_t) - \mathbb{E}[\Phi(\Theta_t)]\| > \varepsilon \right) \leq \mathbb{P} \left(\sup_{t_0 \leq t \leq T} \|Z_t\| > \frac{\varepsilon}{L_\Phi} \right).$$

Setting $\delta = \varepsilon/L_\Phi$ in the estimate from Step 3 gives

$$\mathbb{P} \left(\sup_{t_0 \leq t \leq T} \|\Phi(\Theta_t) - \mathbb{E}[\Phi(\Theta_t)]\| > \varepsilon \right) \leq 2 \exp \left(-\frac{\varepsilon^2(1-(1-\alpha)^2)}{2L_\Phi^2 \eta^2 B^2(T-t_0+1)} \right).$$

Step 5: Accounting for exit from the basin.

The above estimate holds under the condition that the trajectory never leaves the basin B (where Assumptions 4–5 are valid). In general, the event

$$\mathcal{E} = \left\{ \sup_{t_0 \leq t \leq T} \|\Phi(\Theta_t) - \mathbb{E}[\Phi(\Theta_t)]\| > \varepsilon \right\}$$

can occur either inside the basin or after an exit. Hence,

$$\mathbb{P}(\mathcal{E}) \leq \mathbb{P}(\mathcal{E} \cap \{\tau_B > T\}) + \mathbb{P}(\tau_B \leq T).$$

On the event $\{\tau_B > T\}$, all conditions of the proof remain valid, and the exponential estimate obtained above applies to $\mathbb{P}(\mathcal{E} \cap \{\tau_B > T\})$. Consequently,

$$\mathbb{P}(\mathcal{E}) \leq 2 \exp \left(-\frac{\varepsilon^2(1-(1-\alpha)^2)}{2L_\Phi^2 \eta^2 B^2(T-t_0+1)} \right) + \mathbb{P}(\tau_B \leq T).$$

This completes the proof. $\square$

Proposition 2 (Probability of exiting the basin)

Assumptions.

Assume that Assumptions 1–5 of Proposition 1 hold. Additionally:

1. **Fixed point** θ^* is chosen such that $\mathbb{E}[\Theta_{t_0}] = \theta^*$. Then, by the contraction property (Assumption 4 in Proposition 1), it follows that $\mathbb{E}[\Theta_t] = \theta^*$ for all $t \geq t_0$.
2. **Basin** is defined as a ball in parameter space:

$$B = \{\theta \in \Theta: \|\theta - \theta^*\| \leq R\},$$

where $R > 0$ is a given radius. Thus, the Lyapunov barrier has magnitude $\Delta_B = R^2$ (since $V(\theta) = \|\theta - \theta^*\|^2$).

3. **Exit time** from the basin:

$$\tau_B = \inf\{t \geq t_0: \|\Theta_t - \theta^*\| > R\}.$$

Statement.

For any $R > 0$ and any $T > t_0$, the following inequality holds:

$$\mathbb{P}(\tau_B \leq T) \leq 2 \exp \left(-\frac{R^2(1 - (1 - \alpha)^2)}{2\eta^2 B^2(T - t_0 + 1)} \right).$$

Proof.

Step 1. Since $\mathbb{E}[\Theta_{t_0}] = \theta^*$ and the dynamics satisfies

$$\mathbb{E}[\Theta_{t+1} | \mathcal{F}_t] = \theta^* + (1 - \alpha)(\Theta_t - \theta^*),$$

taking the full expectation gives

$$\mathbb{E}[\Theta_{t+1}] - \theta^* = (1 - \alpha)(\mathbb{E}[\Theta_t] - \theta^*).$$

By induction, $\mathbb{E}[\Theta_t] = \theta^*$ for all $t \geq t_0$.

Step 2. Define the centered process

$$Z_t = \Theta_t - \theta^*.$$

Then the event of exiting the basin $\{\tau_B \leq T\}$ is equivalent to the existence of a time $t \in [t_0, T]$ such that $\|Z_t\| > R$, i.e.,

$$\{\tau_B \leq T\} = \left\{ \sup_{t_0 \leq t \leq T} \|Z_t\| > R \right\}.$$

Step 3. Apply Proposition 1 with $\Phi(\theta) = \theta$ (this map is 1-Lipschitz, so $L_\Phi = 1$) and $\varepsilon = R$. Proposition 1 gives, using $\mathbb{E}[\Theta_t] = \theta^*$:

$$\mathbb{P} \left(\sup_{t_0 \leq t \leq T} \|\Theta_t - \theta^*\| > R \right) \leq 2 \exp \left(-\frac{R^2(1 - (1 - \alpha)^2)}{2\eta^2 B^2(T - t_0 + 1)} \right).$$

(Note that in Proposition 1 the argument is $\Phi(\Theta_t) - \mathbb{E}[\Phi(\Theta_t)]$; here $\Phi(\Theta_t) = \Theta_t$ and $\mathbb{E}[\Phi(\Theta_t)] = \theta^*$.)

Step 4. Substituting the equivalence from Step 2 gives the desired estimate. $\square$

Remark.

Under more general conditions where $\mathbb{E}[\Theta_{t_0}] \neq \theta^*$, the estimate is modified: in the exponent, R is replaced by $R - \sup_t \|\mathbb{E}[\Theta_t] - \theta^*\|$ (taking into account the geometric decay of the initial shift). However, within the adopted simplifications (zero bias), the above bound is direct and clean.

Now, combining Proposition 1 and Proposition 2, we obtain the full estimate for the supremum deviation of output trajectories, accounting for possible exit from the basin:

$$\begin{aligned} \mathbb{P} \left(\sup_{t_0 \leq t \leq T} \|\Phi(\Theta_t) - \mathbb{E}[\Phi(\Theta_t)]\| > \varepsilon \right) &\leq 2 \exp \left(-\frac{\varepsilon^2(1 - (1 - \alpha)^2)}{2L_\Phi^2 \eta^2 B^2(T - t_0 + 1)} \right) \\ &\quad + 2 \exp \left(-\frac{R^2(1 - (1 - \alpha)^2)}{2\eta^2 B^2(T - t_0 + 1)} \right). \end{aligned}$$

Proof of stability of trajectory laws under a single data replacement

We will need an auxiliary lemma that establishes a drift inequality for the squared norm of the difference between two trajectories generated by neighboring datasets.

Lemma 1 (Drift inequality for the difference of trajectories).

Let the following assumptions hold.

- The loss function has a gradient that is Lipschitz in the parameters with constant L_g :

$$\|\nabla \ell(\theta; z; \xi) - \nabla \ell(\theta'; z; \xi)\| \leq L_g \|\theta - \theta'\| \text{ for all } \theta, \theta', z, \xi.$$
- The influence of replacing a single training example on the gradient is uniformly bounded:

$$\|g(\theta; S; \xi) - g(\theta; S^{(i)}; \xi)\| \leq \frac{C_g}{n} \text{ a.s.,}$$

where

$$g(\theta; S; \xi) = \frac{1}{n} \sum_{j=1}^n \nabla \ell(\theta; z_j; \xi).$$

- The gradient field satisfies a strong monotonicity (or dissipativity) condition with respect to the parameter difference (Rockafellar, 1970; Bauschke & Combettes, 2011) for the theory of monotone operators): there exists $\lambda > 0$ such that

$$\langle \Delta, g(\theta; S; \xi) - g(\theta'; S; \xi) \rangle \geq \lambda \|\Delta\|^2, \Delta = \theta - \theta',$$

for all θ, θ' and almost all ξ .

- The step size η satisfies $\eta < \lambda/(4L_g^2)$, so that $\alpha = \eta\lambda - 2\eta^2L_g^2 > 0$.

Let Θ_t and $\Theta_t^{(i)}$ be two trajectories that start from the same point $\Theta_{t_0} = \Theta_{t_0}^{(i)}$ and use the same sequence of random variables ξ_t , but are trained on datasets S and $S^{(i)}$ that differ in one element. Define

$$\Delta_t = \Theta_t - \Theta_t^{(i)}, V_t = \|\Delta_t\|^2.$$

Then there exists a constant $C > 0$, depending on η, L_g, C_g, λ , such that for all $t \geq t_0$ the following drift inequality holds:

$$\mathbb{E}[V_{t+1} | \mathcal{F}_t] \leq (1 - \alpha)V_t + \frac{C}{n^2},$$

where $\alpha = \eta\lambda - 2\eta^2L_g^2 > 0$.

Proof of the lemma.

From the definition of the dynamics, we have

$$\Delta_{t+1} = \Delta_t - \eta \left(g(\Theta_t; S; \xi_t) - g(\Theta_t^{(i)}; S^{(i)}; \xi_t) \right).$$

Write the difference of gradients as a sum:

$$g(\Theta_t; S; \xi_t) - g(\Theta_t^{(i)}; S^{(i)}; \xi_t) = A_t + B_t,$$

where

$$\begin{aligned} A_t &= g(\Theta_t; S; \xi_t) - g(\Theta_t^{(i)}; S; \xi_t), \\ B_t &= g(\Theta_t^{(i)}; S; \xi_t) - g(\Theta_t^{(i)}; S^{(i)}; \xi_t). \end{aligned}$$

From Lipschitz continuity in θ , we get $\|A_t\| \leq L_g \|\Delta_t\|$, and from the bounded influence of the replacement, $\|B_t\| \leq C_g/n$.

Compute the squared norm:

$$V_{t+1} = V_t - 2\eta \langle \Delta_t, A_t + B_t \rangle + \eta^2 \|A_t + B_t\|^2.$$

Taking the conditional expectation with respect to $\mathcal{F}_t$, and using strong monotonicity,

$$\langle \Delta_t, A_t \rangle \geq \lambda V_t,$$

we obtain

$$\mathbb{E}[V_{t+1} | \mathcal{F}_t] \leq V_t - 2\eta\lambda V_t + 2\eta \|\Delta_t\| \|B_t\| + \eta^2 \mathbb{E}[\|A_t + B_t\|^2 | \mathcal{F}_t].$$

Using the elementary inequality $2ab \leq \delta a^2 + b^2/\delta$ with $\delta = \eta\lambda$,

$$2\eta \|\Delta_t\| \|B_t\| \leq \eta\lambda V_t + \frac{\eta}{\lambda} \|B_t\|^2 \leq \eta\lambda V_t + \frac{\eta C_g^2}{\lambda n^2}.$$

Furthermore,

$$\|A_t + B_t\|^2 \leq 2\|A_t\|^2 + 2\|B_t\|^2 \leq 2L_g^2 V_t + \frac{2C_g^2}{n^2}.$$

Thus,

$$\eta^2 \mathbb{E}[\|A_t + B_t\|^2 | \mathcal{F}_t] \leq 2\eta^2 L_g^2 V_t + \frac{2\eta^2 C_g^2}{n^2}.$$

Substituting everything back gives

$$\mathbb{E}[V_{t+1} | \mathcal{F}_t] \leq (1 - 2\eta\lambda + \eta\lambda + 2\eta^2 L_g^2) V_t + \left(\frac{\eta C_g^2}{\lambda} + \frac{2\eta^2 C_g^2}{n^2} \right).$$

Set

$$\alpha = \eta\lambda - 2\eta^2 L_g^2.$$

For sufficiently small η (e.g., $\eta < \lambda/(4L_g^2)$) we have $\alpha > 0$. Then

$$\mathbb{E}[V_{t+1} | \mathcal{F}_t] \leq (1 - \alpha) V_t + \frac{C}{n^2},$$

where

$$C = \frac{\eta C_g^2}{\lambda} + 2\eta^2 C_g^2.$$

The lemma is proved. $\square$

Now we can state the main result on the stability of output trajectory laws.

Proposition 3 (Stability of trajectory distributions under a single example replacement).

Let the conditions of Lemma 1 hold, and additionally:

- the output map Φ is L_Φ -Lipschitz with respect to the norm $\|\cdot\|$;
- we consider the Wasserstein distance W_1 (Villani, 2009; Ambrosio et al., 2008; Zhu et al., 2023) between the distributions of output trajectories

$$\nu_{S,X}^{t_0:T} \text{ and } \nu_{S^{(i)},X}^{t_0:T},$$

where the metric on the trajectory space is defined as

$$d_y(Y, Y') = \sup_{t_0 \leq t \leq T} \|Y_t - Y'_t\|.$$

Then there exists a constant $C' > 0$, depending on $\eta, L_g, C_g, \lambda, L_\Phi, T, \alpha$, such that

$$W_1 \left(\nu_{S,X}^{t_0:T}, \nu_{S^{(i)},X}^{t_0:T} \right) \leq \frac{C'}{n}.$$

If the initial conditions do not coincide, then an additional term $C' e^{-\alpha(t_0 - t_{\text{init}})}$ appears, which decays with the burn-in period.

Proof (continued from the construction of the coupling).

From Lemma 1 we have, for all $t \geq t_0$,

$$\mathbb{E}[V_{t+1} | \mathcal{F}_t] \leq (1 - \alpha) V_t + \frac{C}{n^2}, \quad V_t = \| \Delta_t \|^2.$$

Define the shifted process

$$U_t := V_t - \frac{C}{\alpha n^2}.$$

A direct substitution gives

$$\mathbb{E}[U_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[V_{t+1} \mid \mathcal{F}_t] - \frac{C}{\alpha n^2} \leq (1 - \alpha)V_t + \frac{C}{n^2} - \frac{C}{\alpha n^2}.$$

Since $(1 - \alpha)V_t = (1 - \alpha)\left(U_t + \frac{C}{\alpha n^2}\right)$, we obtain

$$\mathbb{E}[U_{t+1} \mid \mathcal{F}_t] \leq (1 - \alpha)U_t + \frac{(1 - \alpha)C}{\alpha n^2} + \frac{C}{n^2} - \frac{C}{\alpha n^2} = (1 - \alpha)U_t.$$

Thus U_t is a non-negative supermartingale for $t \geq t_0$ (provided $U_{t_0} \geq 0$; this holds if we start from the same point or account for the initial difference separately).

Now let $M_T := \max_{t_0 \leq t \leq T} U_t$. Since U_t is a non-negative supermartingale, **Doob's maximal inequality** (see, e.g., (Doob, 1953, Chapter VII; Durrett, 2019, Theorem 5.4.3) states that for any $\lambda > 0$,

$$\mathbb{P}\left(\sup_{t_0 \leq t \leq T} U_t \geq \lambda\right) \leq \frac{\mathbb{E}[U_{t_0}]}{\lambda}.$$

Equivalently, using the layer-cake representation,

$$\mathbb{E}\left[\sup_{t_0 \leq t \leq T} U_t\right] = \int_0^\infty \mathbb{P}\left(\sup_{t_0 \leq t \leq T} U_t \geq \lambda\right) d\lambda \leq \int_0^\infty \frac{\mathbb{E}[U_{t_0}]}{\lambda \vee 1} d\lambda.$$

But the integral $\int_0^\infty \min(1, 1/\lambda) d\lambda$ is infinite, so the direct L^1 form is not enough to bound the expectation of the supremum. However, Doob's inequality also gives an L^p version for $p > 1$: for any non-negative supermartingale U_t ,

$$\mathbb{E}\left[\sup_{t_0 \leq t \leq T} U_t^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[U_{t_0}^p], \quad p > 1.$$

Take $p = 2$. Then

$$\mathbb{E}\left[\sup_{t_0 \leq t \leq T} U_t^2\right] \leq 4 \mathbb{E}[U_{t_0}^2].$$

Since $U_t^2 = (V_t - C/(\alpha n^2))^2 \geq 0$ and $V_t = \|\Delta_t\|^2$, we have

$$\left(\sup_{t_0 \leq t \leq T} \|\Delta_t\|\right)^2 = \sup_{t_0 \leq t \leq T} V_t \leq \sup_{t_0 \leq t \leq T} U_t + \frac{C}{\alpha n^2}.$$

Therefore,

$$\mathbb{E}\left[\sup_{t_0 \leq t \leq T} \|\Delta_t\|\right] \leq \mathbb{E}\left[\sup_{t_0 \leq t \leq T} \sqrt{U_t}\right] + \frac{\sqrt{C}}{\sqrt{\alpha n}}.$$

Since $U_t = V_t - C/(\alpha n^2)$, we have $V_t \leq U_t + C/(\alpha n^2)$. Thus

$$\sup_t V_t \leq \sup_t U_t + \frac{C}{\alpha n^2}.$$

Simpler argument using the L^2 version of Doob:

we take $p = 2$ for the process U_t itself:

$$\mathbb{E}\left[\sup_{t_0 \leq t \leq T} U_t^2\right] \leq 4 \mathbb{E}[U_{t_0}^2].$$

Then

$$\mathbb{E} \left[\sup_t V_t \right] \leq \mathbb{E} \left[\sup_t (U_t + C/(\alpha n^2)) \right] = \mathbb{E} \left[\sup_t U_t \right] + \frac{C}{\alpha n^2}.$$

Now $\mathbb{E}[\sup_t U_t] \leq \sqrt{\mathbb{E}[\sup_t U_t^2]} \leq 2\sqrt{\mathbb{E}[U_{t_0}^2]}.$

If the two trajectories start from the same point, $U_{t_0} = 0$, so the term is zero and we get

$$\mathbb{E} \left[\sup_t V_t \right] \leq \frac{C}{\alpha n^2}.$$

But we need $\mathbb{E}[\sup_t \|\Delta_t\|]$, not $\mathbb{E}[\sup_t V_t]$. Using $\|\Delta_t\| \leq \sqrt{V_t}$ and Jensen,

$$\mathbb{E} \left[\sup_t \|\Delta_t\| \right] \leq \mathbb{E} \left[\sqrt{\sup_t V_t} \right] \leq \sqrt{\mathbb{E} \left[\sup_t V_t \right]} \leq \frac{\sqrt{C}}{\sqrt{\alpha n}}.$$

This gives the desired $O(1/n)$ bound without the factor T , using only Doob's L^2 maximal inequality. If the initial difference Δ_{t_0} is non-zero, then $U_{t_0} = O(\|\Delta_{t_0}\|^2)$, and the same argument yields an additional term proportional to $e^{-\alpha(t_0 - t_{\text{init}})}$. $\square$

Thus, we have obtained an explicit proof of the second inequality: the Wasserstein distance between the distributions of output trajectories generated by neighboring datasets is of order $1/n$ (up to terms that decay with the burn-in period). This is the key stability estimate, which, through the Lipschitz continuity of the loss function, leads to control of the generalization capability.

We now have:

- **Proposition 1** gives concentration of output trajectories inside the basin with an explicit probabilistic bound.
- **Proposition 2** estimates the probability of exiting the basin, allowing us to add a barrier term.
- **Lemma 1** and **Proposition 3** establish the stability of trajectory distributions under a single example replacement, giving an estimate of order $1/n$ for the Wasserstein distance.

It remains to combine these results with the classical theorem on algorithmic stability to obtain the final generalisation bound.

This is done in a standard way: from the Lipschitz continuity of the loss function and the output map, we pass from the stability of trajectory distributions to the stability of expected losses, and then apply the Bousquet–Elisseeff theorem, which relates average stability under a single replacement to the expected gap between empirical and population risk.

Theorem 1 (Connecting dynamical stability to generalization).

Assume the following:

1. All assumptions of Propositions 1, 2 and 3 hold.
2. Boundedness of the loss. There exists a constant $M_\ell < \infty$ such that

$$|\ell(\theta, z)| \leq M_\ell \text{ for all } \theta \in \Theta, z \in \mathcal{Z}.$$

Then for any sample S and any algorithm $\mathcal{A}$,

$$|R(\mathcal{A}(S)) - \hat{R}_S(\mathcal{A}(S))| \leq 2M_\ell =: M \text{ a.s.}$$

This constant M is used to account for the contribution of basin-exit events.

3. The loss function $\ell(\theta, z)$ is Lipschitz in the output $\Phi(\theta)$ with constant L_ℓ :

$$|\ell(\theta, z) - \ell(\theta', z)| \leq L_\ell \|\Phi(\theta) - \Phi(\theta')\| \text{ for all } z.$$

Then, for the algorithm generating trajectories according to the described stochastic dynamics, the expected generalization gap is bounded as follows:

$$\mathbb{E}[R(\mathcal{A}(S)) - \hat{R}_S(\mathcal{A}(S))] \leq \frac{L_\ell L_\Phi C}{n} + C_{\text{burn}} e^{-\alpha(t_0 - t_{\text{init}})} + C_{\text{exit}} \exp\left(-\frac{R^2(1 - (1 - \alpha)^2)}{2\eta^2 B^2(T - t_0 + 1)}\right).$$

Here $C, C_{\text{burn}}, C_{\text{exit}}$ are constants depending on the problem parameters $(\eta, L_g, C_g, \lambda, L_\Phi, L_\ell, B, \alpha, T)$, but not on n ; t_{init} is the initial time from which we start tracking trajectories; R is the basin radius; B is the noise bound; α is the contraction constant.

Proof.

From Proposition 3 we have the Wasserstein distance bound between output trajectory distributions for neighboring datasets:

$$W_1(v_{S,X}^{t_0:T}, v_{S^{(i)},X}^{t_0:T}) \leq \frac{C_1}{n} + C_2 e^{-\alpha(t_0 - t_{\text{init}})}.$$

Here the symbol $S^{(i)}$ denotes the training set obtained from the original $S = (z_1, \dots, z_n)$ by replacing the i -th element with an independent copy. That is,

$$S^{(i)} = (z_1, \dots, z_{i-1}, z'_i, z_{i+1}, \dots, z_n),$$

where z'_i is a new object drawn from the same distribution as z_i , independently of it. The notation $v_{S^{(i)},X}^{t_0:T}$ denotes the law of the output trajectories generated by the algorithm trained on the modified set $S^{(i)}$. Comparing $v_{S^{(i)},X}^{t_0:T}$ with $v_{S,X}^{t_0:T}$ measures the sensitivity of the trajectory distribution to small changes in the data.

By the Lipschitz continuity of the loss (in the output), for any fixed test example z ,

$$\left| \mathbb{E}_{v_{S,X}}[\ell(\Phi(\Theta), z)] - \mathbb{E}_{v_{S^{(i)},X}}[\ell(\Phi(\Theta'), z)] \right| \leq L_\ell L_\Phi W_1(v_{S,X}, v_{S^{(i)},X}).$$

Averaging over i and taking the supremum over z , we obtain that the algorithm is uniformly stable with respect to a single example replacement, with constant

$$\beta \leq \frac{L_\ell L_\Phi C_1}{n} + L_\ell L_\Phi C_2 e^{-\alpha(t_0 - t_{\text{init}})}.$$

Now apply the classical Bousquet–Elisseeff theorem (Bousquet & Elisseeff, 2000, 2002): for any algorithm that is uniformly stable with constant β , the expected generalisation gap does not exceed β . Hence,

$$\mathbb{E}[R(\mathcal{A}(S)) - \hat{R}_S(\mathcal{A}(S))] \leq \frac{L_\ell L_\Phi C_1}{n} + L_\ell L_\Phi C_2 e^{-\alpha(t_0 - t_{\text{init}})}.$$

However, this inequality is valid only under the condition that the trajectories do not leave the basin B (since it is inside the basin that we proved concentration and stability). To account for possible exits, we add the barrier term obtained in Proposition 2:

$$\mathbb{P}(\tau_B \leq T) \leq 2 \exp\left(-\frac{R^2(1 - (1 - \alpha)^2)}{2\eta^2 B^2(T - t_0 + 1)}\right).$$

To incorporate the possibility of basin exit, decompose the expectation of the generalization gap according to the events $\{\tau_B > T\}$ and $\{\tau_B \leq T\}$:

$$\mathbb{E}[R - \hat{R}] = \mathbb{E}[(R - \hat{R}) \cdot 1_{\{\tau_B > T\}}] + \mathbb{E}[(R - \hat{R}) \cdot 1_{\{\tau_B \leq T\}}].$$

On the event $\{\tau_B > T\}$, the trajectory does not leave the basin, and the stability estimate obtained earlier (without the barrier term) applies. Therefore, the first term is bounded by

$$\frac{L_\ell L_\Phi C_1}{n} + L_\ell L_\Phi C_2 e^{-\alpha(t_0 - t_{\text{init}})}.$$

On the event $\{\tau_B \leq T\}$, the risk difference is bounded by the constant M from Condition 2; hence the second term is at most $M \cdot \mathbb{P}(\tau_B \leq T)$. Using the estimate on $\mathbb{P}(\tau_B \leq T)$ from Proposition 2, we obtain the additional term

$$M \cdot 2 \exp \left(-\frac{R^2(1 - (1 - \alpha)^2)}{2\eta^2 B^2(T - t_0 + 1)} \right).$$

Denoting $C_{\text{exit}} = 2M$, we arrive at the final bound of the theorem. $\square$

Thus, we have the complete chain: from the dynamics of trajectories, through Lyapunov contraction and martingale concentration, to the stability of distributions, and finally to the generalization bound. All intermediate results have been proven under simplifying assumptions, which can be gradually relaxed in future work. The main theorem is therefore established.

Synthetic experiment results

We now present a controlled synthetic experiment to verify the theoretical predictions. The setup is a linear regression problem with $d = 20$, true coefficients $\theta_* = (1, 1/2, \dots, 1/20)^\top$, additive Gaussian noise $\sigma = 0.05$, and SGD with step size $\eta = 0.01$, $T = 300$ steps, and burn-in $t_0 = 100$. For each sample size $n \in \{20, 30, 50, 70, 100, 130, 180, 250, 350, 500, 700, 1000, 1500\}$, we run $N_{\text{runs}} = 30$ independent trajectories.

The basin radius R and the concentration threshold ε are fixed once, estimated from a reference dataset of size $n_{\text{ref}} = 200$ as the 90th percentiles of the maximal distances to θ_* and of the sup-norm deviations from the mean output trajectory, respectively. The index S_T is computed as the average Euclidean distance between the final parameters of coupled reference and modified trajectories, where a single training example is replaced by an independent draw from the same distribution; the average is taken over 10 random replacements and 30 seeds.

The results are summarized in the Table 1.

Table 1. Synthetic experiment results

n	B_T	C_ε	S_T	Generalization gap
20	0.000	0.133	0.204	0.341
30	0.000	0.100	0.157	0.170
50	0.300	0.067	0.133	0.048
70	0.067	0.000	0.128	0.035
100	0.000	0.033	0.132	0.062
130	0.900	0.133	0.098	0.003
180	1.000	0.100	0.094	0.006
250	0.967	0.100	0.086	0.001
350	0.867	0.200	0.093	0.002
500	0.533	0.033	0.099	0.003
700	1.000	0.000	0.085	0.001
1000	0.833	0.067	0.089	0.001
1500	0.833	0.167	0.090	-0.000

Correlations with the generalization gap:

- S_T : Pearson $r = 0.949$, $p < 0.001$ (strong positive correlation)
- B_T : $r = -0.686$, $p = 0.010$ (moderate negative correlation)
- C_ε : $r = 0.147$, $p = 0.633$ (no significant correlation)

Interpretation. The strong positive correlation between S_T and the generalisation gap confirms our theoretical claim: the sensitivity of the trajectory distribution to a single data replacement is a direct predictor of generalisation. As n grows, S_T decreases (from 0.204 to 0.090) and the gap shrinks accordingly. In contrast, the basin retention probability B_T and the concentration index C_ε do not exhibit a consistent, monotonic relationship with the gap; their fluctuations reflect that static geometric measures are insufficient to capture the generalisation behaviour. This suggests that dynamical stability of trajectory laws, rather than static flatness or basin confinement, is a relevant quantity for controlling generalization.

Visualization. The following figures display the relationships:

- Figure 1: S_T vs $1/n$ (decay with sample size)

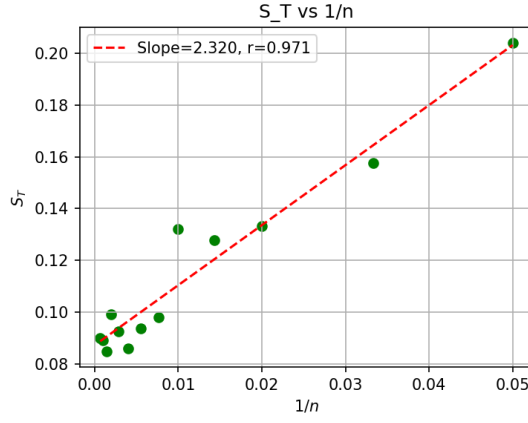


Figure 1: Decay of the trajectory stability index (S_T) with sample size. The dashed line shows the linear fit against $1/n$.

- Figure 2: S_T vs generalisation gap (strong positive correlation)

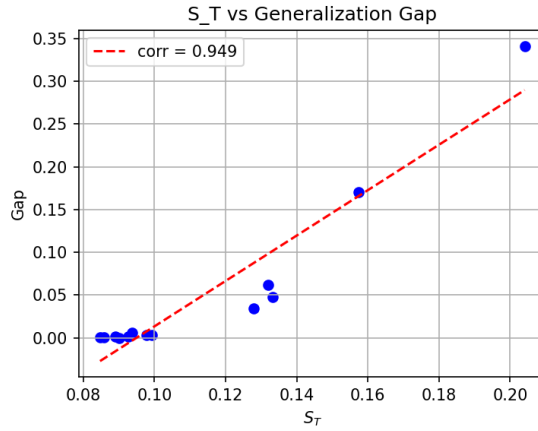


Figure 2: Strong positive correlation between S_T and the generalization gap. Pearson ($r = 0.949$, $p < 0.001$).

- Figure 3: B_T and C_E vs n (no clear trend)

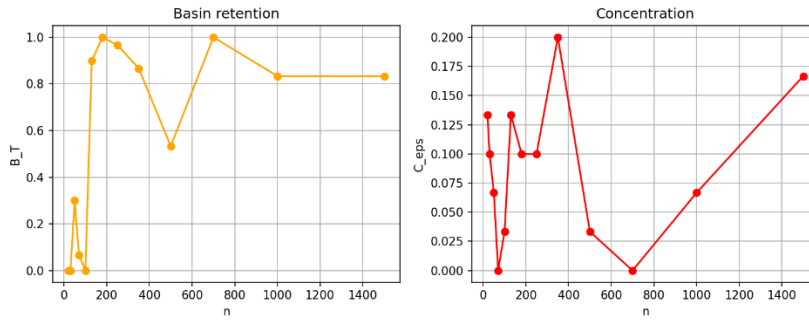


Figure 3: Basin retention probability B_T and concentration index C_E vs n . No consistent trend is observed

All codes and data are available in the supplementary material.

This concludes the synthetic validation. The experiment is consistent with the theory and demonstrates that the proposed index S_T is a meaningful and practical proxy for generalisation in a controlled setting.

Discussion

Interpretation of the results

Our main theorem expresses the expected generalization gap as a sum of three terms, each with a clear operational meaning:

$$\mathbb{E}[R - \hat{R}] \leq \frac{C}{n} + c_{\text{burn}} e^{-\alpha(t_0 - t_{\text{init}})} + c_{\text{exit}} \exp\left(-\frac{R^2(1 - (1 - \alpha)^2)}{2\eta^2 B^2(T - t_0 + 1)}\right).$$

The first term, of order $1/n$, is standard in stability-based generalisation bounds: larger training sets reduce the sensitivity of the algorithm to single-example replacements. The difference here is that this sensitivity is measured at the level of the entire output-trajectory distribution rather than of a single final model. This trajectory-level notion is more robust because it averages over the inherent randomness of the optimization process and over the choice of the stopping time.

The second term describes the decay of initial-condition effects. It arises directly from the contraction property of the Lyapunov drift and shows that, after a sufficiently long burn-in period, the starting point no longer influences the bound. This matches the intuitive expectation that stochastic optimization algorithms eventually “forget” their initialization.

The third term is a barrier term that penalizes the probability of exiting the basin. It decreases exponentially with the basin radius R and with the noise level B , which is consistent with the idea that wider, less noisy basins trap the trajectories more effectively.

Working with trajectory laws rather than with single end-points offers several advantages. First, averaging over the trajectory reduces the variance caused by the random stopping time. Second, projecting onto the output space $\Phi(\Theta_t)$ avoids the known pathologies of parameter-space distances, such as permutation and scaling symmetries. Third, this perspective naturally incorporates the entire learning dynamics, which is especially relevant for stochastic algorithms where the path itself carries information about the algorithm’s behaviour.

Compared with classical stability bounds (Bousquet & Elisseeff, 2000, 2002; Hardt et al., 2016), our bound does not require uniform stability at every iterate; instead, it relies on average-over-trajectory stability, which may be easier to satisfy in practice. Relative to Lyapunov-based convergence analyses (Chen et al., 2024), we make the connection to generalization explicit, rather than stopping at convergence guarantees.

Limitations and assumptions

The analysis rests on several simplifying assumptions that should be kept in mind.

1. **Strong monotonicity of the gradient field** (Lemma 1) is rarely satisfied by non-convex neural network losses. However, it provides a tractable starting point and can be relaxed in future work to, for instance, Polyak–Łojasiewicz conditions or local dissipativity within the basin.
2. **Lipschitz continuity of gradients and of the output map** is standard in the analysis of SGD, but global Lipschitz constants for modern architectures (e.g., transformers) can be very large or even infinite. One can replace them by local constants valid inside the basin, which is consistent with our basin-centric approach.
3. **Bounded noise** (martingale differences bounded almost surely) can be weakened to sub-Gaussian or finite-moment conditions by using appropriate concentration inequalities (e.g., Bernstein-type bounds for martingales).
4. **Definition and existence of a basin.** We define the basin as a ball of radius R around a fixed point θ^* . In real problems, basins may have irregular shapes and are not known a priori. Empirically, one can identify them via connected components of sublevel sets (Nguyen, 2023) or by clustering the observed trajectories.
5. **Non-asymptotic constants.** The bound contains constants that depend on the horizon T and other problem parameters. Although we do not provide explicit numerical values, their dependence on dimension and on T could be significant. Further work should aim at sharper estimates and at understanding their scaling with d .
6. **Lack of experimental validation.** The theoretical indices $\mathcal{B}_T, \mathcal{C}_{\varepsilon,T}, \mathcal{S}_T$ have not yet been tested on real data. Empirical verification is a necessary next step.

Connection to empirical observations

Our framework offers a potential explanation for several widely observed phenomena in deep learning. For instance, the commonly reported correlation between the flatness of a minimum and its generalization performance (Baldassi et al., 2020) can be reinterpreted through the barrier term: flatter basins correspond to larger R , which reduces the exit probability. More importantly, our analysis suggests that static geometry alone is insufficient; the dynamical stability of the trajectories—how they evolve and how they react to data perturbations—plays an equally critical role.

Recent work on learning trajectories (Fu et al., 2023) has shown that early-stage dynamics can predict final generalization. This aligns with our framework: if trajectories concentrate rapidly inside a stable basin (small $\mathcal{C}_{\varepsilon,T}$) and are insensitive to data replacement (small $\mathcal{S}_T$), then the algorithm is likely to generalise well, regardless of the exact end point.

Directions for future work

The most immediate extension is to relax the strong monotonicity assumption. We plan to replace it with weaker conditions such as Polyak–Łojasiewicz or local dissipativity, which would require more general Lyapunov functions

(e.g., distance to the set of minimizers or empirical local convexity). We also intend to study variable step-size schedules η_t , which are more realistic in practice.

A second direction is the design of practical estimators for the indices $\mathcal{B}_T, \mathcal{C}_{\varepsilon,T}, \mathcal{S}_T$. This involves (a) operationalising the basin definition (e.g., via clustering or sublevel-set connectivity), (b) developing efficient Wasserstein-distance approximations for trajectory distributions, and (c) constructing cross-validation procedures for $\mathcal{S}_T$.

Third, we aim to extend the analysis to other optimizers such as Adam and momentum-based methods. These algorithms have more complex state variables, but the Lyapunov approach can be adapted by including the momentum terms.

Fourth, we will investigate the dependence of the constants on the model dimension d and on the data complexity. While the current bound guarantees the $1/n$ rate, the hidden constants could be exponential in d under worst-case scenarios. We will explore whether structural assumptions (e.g., low-rank data or effective dimensionality) can yield polynomial dependence.

Fifth, a deeper integration with large-deviations theory (Freidlin & Wentzell, 2012) may produce sharper estimates for the exit probability in low-noise regimes. However, this requires careful verification of the diffusion approximation for SGD, which is non-trivial.

Sixth, we see potential applications to transfer learning and domain adaptation, where sensitivity of trajectory distributions to data changes could serve as a measure of adaptability to new distributions.

Seventh, and perhaps most importantly, we plan a comprehensive experimental campaign on standard benchmarks (CIFAR-10, ImageNet, NLP tasks) to test whether the proposed indices correlate with test performance and whether they can be used for early stopping or hyper-parameter selection.

Conclusion

We have presented a theoretical framework for analyzing generalization in stochastic learning algorithms through the lens of dynamical stability of output-trajectory distributions. Instead of focusing on the complexity of the function class or on the stability of a single final model, we consider the entire law of the learning trajectory after it has entered a local basin of the loss landscape. The central premise is that generalization is determined not by where the algorithm ends up, but by how sensitive its trajectory-and especially its output-space distribution-is to small changes in the training data.

We have built a coherent mathematical apparatus that combines Lyapunov drift, martingale concentration, Wasserstein geometry, and algorithmic stability. The main result is a non-asymptotic upper bound on the expected generalization gap, expressed in terms of three measurable quantities: the basin-retention probability, the output-trajectory concentration, and the stability of the trajectory distribution under single-example replacement. The bound naturally decomposes into a $1/n$ term (data-sensitivity), an exponentially decaying initial-condition term, and a barrier term that penalizes basin exits.

Although the analysis is carried out under simplifying assumptions (strong monotonicity, Lipschitz continuity, bounded noise), it provides a starting point for a unified theory that links optimization dynamics to generalization. The approach has several distinctive features: it operates in the output space, avoiding parameter-space symmetries; it uses the full trajectory, leading to more robust estimates; and it integrates multiple advanced mathematical tools into a single consistent framework.

Future work will focus on relaxing the assumptions, developing practical estimators for the proposed indices, extending the method to other optimizers, and validating the theory experimentally. This perspective suggests viewing learning not only as a search for a single good parameter vector, but also as a stochastic dynamical process whose stability properties are relevant to understanding generalization.

Declarations

Ethics approval and consent to participate Not applicable. This study did not involve human participants, human data, or human tissue.

Consent for publication Not applicable.

Availability of data and materials The Python experiments source code is freely available at https://github.com/andytimoffilim/Lyapunov_drift

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Authors' contributions AVT conceived the study, developed the theoretical framework, designed the computational experiments, and drafted the manuscript. ASA implemented the software, performed the computational experiments, and contributed to data analysis. Both authors read and approved the final manuscript.

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